

Amrita School of Engineering, Bangalore-35

23MAT117-Linear Algebra

Lab Session Sheet – 07

QR Decompositions

❖ **QR Decompositions using MATLAB**

$[Q, R] = \text{qr}(A, 0)$ % QR decomposition of matrix A.

➤ **QR Decomposition for finding eigenvalues of matrices numerically:**

```
A=randi(10,2,2);  
eig(A) B=A;  
for i= 1:10  
    [Q R] = qr(B,0);    % B=QR.  
    B=R*Q;  
    % Creating a new B=RQ and repeat the computation. B=RQ retains  
    eigenvalues of A (as RQ is similar to A) but approaches to an upper triangular  
    matrix. That is, diagonal elements of B approaches to eigenvalues of A.  
end  
diag(B)    % Note that eig(A) and diag(B) are almost the same values.
```

Practice questions

1. Using QR decomposition find the eigenvalues of a random integer symmetric matrix of order 15 and verify the result using direct evaluation of eigenvalues.
2. Generate a random integer 7 by 7 matrix A of rank 7.
 - a) Verify the rank of A is 7.
 - b) Find the QR decomposition of A.
3. Using QR decomposition find the eigenvalues of the following matrices using 50 iterations in MATLAB.

a) $A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 3 \\ 5 & 7 & 0 \end{bmatrix}$

$$\text{b) } B = \begin{bmatrix} 1 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 2 \end{bmatrix}$$

c) A random 50 by 50 matrix.

$$\text{d) } C = \begin{bmatrix} 1 & 0 & 1 & 0 \\ -1 & 1 & 1 & -1 \\ 1 & 0 & 1 & 0 \\ -1 & 1 & 1 & -1 \end{bmatrix}$$

e) A random 5 by 5 matrix of rank 3.

4. Generate a random integer 12 by 12 matrix A of rank 7.

a) Verify the rank of A is 7.

b) Find the QR decomposition of A.